Proactive Social Compliance Modeling using Agentic Theory of Mind

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***Abstract*—Autonomous vehicles (AVs) typically react to pedestrians only after crossing behavior becomes apparent. This paper aims to develop a neuro-symbolic pipeline that proactively infers pedestrian mental states and uses these predictions for explainable AV decision-making. Pose landmarks are extracted from the Pedestrian Intention Estimation (PIE) dataset using MediaPipe and transformed into a 36-dimensional per-frame feature representation, which is aggregated into 30-frame sliding-window sequences. A six-class mental-state taxonomy—Waiting, Hesitant, Committed, Distracted, Aggressive/Abnormal Gait, and Jaywalk—is generated using a deterministic rule-based labeling procedure grounded in PIE annotations and observable pedestrian behavior. Two temporal deep-learning classifiers, a two-layer Long Short-Term Memory (LSTM) network and a two-layer Gated Recurrent Unit (GRU) network, are trained for mental-state inference, with an equal-weight ensemble used to combine their predictions. The inferred mental state is integrated with an image-space time-to-collision (TTC) estimate within a transparent rule-based decision agent that maps pedestrian risk to proactive AV actions. The decision mechanism is further extended to handle multiple pedestrians by considering the most critical pedestrian risk. Evaluation on held-out PIE sequences demonstrates the feasibility of the proposed approach for distinguishing different pedestrian behavioral patterns, while challenges remain in separating minority and behaviorally overlapping classes. The resulting neuro-symbolic framework provides an interpretable link between pedestrian state inference, risk estimation, and AV decision-making.**

***Index Terms*—pedestrian intention prediction, Theory of Mind, LSTM, GRU, time-to-collision, autonomous vehicles, PIE dataset**

I. Introduction

Most deployed autonomous-vehicle (AV) perception stacks treat pedestrians re-actively: a trajectory-prediction module extrapolates future position from past motion, and the planner responds once a crossing has effectively begun. This does not distinguish a pedestrian who is waiting at the curb from one who is hesitant, distracted, or about to jaywalk - cases that call for different anticipatory responses. Pedestrians face particularly elevated risks at crossings without traffic signals, where interactions are often informally negotiated. This motivates systems that reason about pedestrian behavioral states rather than relying solely on positional information.

Theory of Mind (ToM) - inferring another agent's beliefs, intentions, and desires from observable behavior - offers a useful framing: instead of "where will this pedestrian be," the system asks "what is this pedestrian's current behavioral state." This is empirically grounded: Goodhew et al. [2] link individual ToM ability to hazard-perception skill in drivers, and Nori et al. [3] link cognitive ToM to more accurate risk assessment. Building on this perspective, the proposed approach is framed as an agentic perceive-reason–act–explain system that infers pedestrian behavioral states, plans proactive vehicle responses, and provides natural-language explanations

This paper reports only the components that were implemented and evaluated: pose-based feature extraction, rule-based mental-state labeling, two temporal classifiers, an equal-weight ensemble, and a rule-based decision agent extended to multiple pedestrians. Contributions: (i) a corrected and reproducible 36-feature extraction pipeline for the PIE dataset, addressing two identified implementation defects; (ii) a rule-based six-class labeling procedure grounded in PIE annotations, with statistical sanity checks.; (iii) a benchmark of two temporal classifiers using identical data splits, together with an equal-weight ensemble; (iv)a transparent, TTC-based decision agent that generates natural-language justifications and supports multiple pedestrians.; and (v) a transparent comparison of the obtained results with the project's original design targets and relevant published results

II. Related Work

Pedestrian intention prediction is well studied, and PIE [1] is a widely used benchmark. Surveys [4]–[6] note that many existing systems formulate the task as binary crossing/non-crossing prediction, potentially overlooking intermediate be-

havioral states such as hesitation. This limitation motivates the six-class behavioral-state taxonomy investigated in this paper.

Methodologically closest to this work are pose-based recurrent pipelines. Li et al. [9] estimate pedestrian motion states from 2D pose using a GRU, while Ahmed et al. [10] use 3D joint representations with an LSTM. Dasgupta and Saha [11] compare LSTM, GRU, and 1D-CNN architectures in a MediaPipe-based pipeline, reporting higher accuracy for GRU than LSTM; this comparison is revisited in the experimental results. More complex architectures have also reported higher performance at increased computational complexity. Xie et al. [12] achieve a precision of 92% on PIE using a graph-embedded transformer, while Yesilyurt and Woda [13] report 98.3% using CNN backbones with cost-sensitive learning for a binary classification task. These results exceed those obtained in this study, potentially reflecting differences in input representations, model complexity, and task formulation.

ToM-based frameworks for road-user behaviour include StylePredict [14] and a thesis that combines policy graphs with ToM for explainable driving [15]. Explainability has also been explored through SAFEXPLAIN [16] and by combining intent prediction with textual explanations [17], with the latter reporting improved predictive performance. These studies support the inclusion of natural-language justifications alongside the decisions generated by the proposed agent. Varga et al. [18] map TTC to a continuous crossing likelihood for model predictive control (MPC) safety costs, providing a conceptual reference for the discretized TTC-based decision mechanism used in this work.

Two further research strands help define the scope of this study. McAllister et al. [19] show that likelihood-based training objectives can under-represent rare, safety-critical behaviors such as jaywalking. Al-Shareeda et al. [20] evaluate a closed-loop PPO-LSTM agent with hesitant pedestrians and report a 5.84% collision rate; such closed-loop safety evaluation is beyond the scope of the present study. Rasouli and Kotseruba [21] distinguish intention estimation, action prediction, and risk assessment. The present work is primarily aligned with intention estimation, while additionally using the inferred behavioral states to support rule-based risk-aware decision making.

III. Dataset and Problem Formulation

This work uses the PIE dataset [1], comprising six annotated sets of dashcam videos with per-frame pedestrian attributes, including action, crossing status, gaze direction, and gesture, as well as track-level intention and temporal annotations. Separate bounding-box tracks for traffic lights, crosswalks, and vehicles provide contextual information.

The task consists of two stages. First, given a trailing 30-frame window (approximately 1 s at 30 fps) of 36-dimensional per-frame features, the system predicts one of six pedestrian behavioural states: Waiting, Hesitant, Committed, Distracted, Aggressive/Abnormal Gait, and Jaywalk. Second, given the predicted state, current-frame features, and the states of other detected pedestrians, the decision agent selects a discrete AV action for each frame. This task relates to, but is distinct from, the intention-estimation, action-prediction, and risk-assessment tasks described by Rasouli and Kotseruba [21]. It is primarily aligned with intention estimation while extending the prediction target from binary crossing behaviour to six behavioural states.

TABLE I
Extracted Features for Pedestrian State Modeling

Group	#	Feature
Motion	1	Current speed
	2	Average speed
	3	Maximum speed
	4	Acceleration
	5	Deceleration
	6	Speed variance
	7	Step frequency
	8	Left step length
	9	Right step length
	10	Pause between steps
Pose	11	Upper-body angle
	12	Lower-body angle
	13	Head angle
	14	Head-turn frequency
	15	Shoulder angle
	16	Hip angle
	17	Left foot angle
	18	Right foot angle
	19	Forward lean
	20	Lateral lean
	21	Body orientation
	22	Body orientation change
Behavioural	23	Pause duration
	24	Hesitation cycles
	25	Total hesitation time
	26	Distance to curb
	27	Distance change rate
	28	Temporal movement probability
	29	Looks left
	30	Looks right
	31	Direction changes
Contextual	32	Traffic-light state
	33	Vehicle distance
	34	Crosswalk presence
	35	Road width
	36	Pedestrian density

IV. Methodology

A. Pose-Based Feature Extraction

MediaPipe Pose [22] extracts 33 body landmarks from each frame. These landmarks are combined with PIE's traffic-light, vehicle, and crosswalk annotations to derive 36 features per frame, organized into four feature groups (Table I).Motion and behavioural features are computed from a trailing 30-frame window, while pose and contextual features capture the pedestrian's current body configuration and surrounding scene conditions. Contextual features include traffic-light state, vehicle distance, crosswalk presence, road width, and pedestrian density.

B. Rule-Based Mental-State Labeling

Mental-state labels are generated deterministically from PIE's frame- and track-level annotations together with the extracted contextual and motion features[23]. Six behavioural states are defined: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. Frames marked as crossing-irrelevant are excluded from labeling. Jaywalk is

assigned when a pedestrian is crossing outside an annotated crosswalk. Within PIE's anticipation window, defined by the track-level `exp_start_point` and `critical_point`, `intention_prob` is used to distinguish Committed (≥ 0.66), Hesitant ($[0.33, 0.66]$), and Distracted (< 0.33 with `look` = not-looking). Outside this window, samples with speed variance above the 75th percentile are assigned to Aggressive, while the remaining samples are assigned to Waiting.

Crossing-irrelevant frames are excluded. The labeling pipeline was further evaluated using statistical sanity checks, including per-feature Kruskal–Wallis tests, a single-feature leakage scan, and an independent K-Means clustering comparison. The leakage analysis identified the expected dependence of the Aggressive class on speed variance, which is directly used in its rule-based definition.

C. Sequence Construction and Data Splits

The extracted frame-level features were organized into 30-frame temporal sequences, corresponding to approximately 1 s at 30 fps. To prevent data leakage, pedestrian tracks were first divided at the track level into training, validation, and test sets using a stratified 70%/15%/15% split with a fixed random seed of 19. Sequence generation was then performed independently within each split, ensuring that frames from the same pedestrian track did not occur across different splits. This produced 17,592 training, 2,340 validation, and 2,791 test sequences.

To improve representation of the under-represented Hesitant class, training tracks containing Hesitant frames were sampled using a stride of one frame, while the remaining tracks used a stride of five frames. Validation and test sequences used the standard five-frame stride. Sequences shorter than 30 frames were left-padded by repeating the earliest available frame to maintain a fixed input length. A `StandardScaler` was fitted using the training sequences only and subsequently applied to the validation and test sets.The resulting class imbalance was addressed using balanced class weights, which were softened using a power transformation to avoid excessive weighting of minority classes. These class weights were combined with the focal-loss objective to further emphasize difficult and under-represented samples during training.

D. Temporal Classifiers: LSTM and GRU

Both models use the same overall architecture, differing only in the recurrent cell type (Table II). Each model receives a 30-frame sequence containing 36 features per frame. The architecture consists of a 128-unit recurrent layer with L2 regularization (10^{-4}), followed by 0.3 dropout, a second 64-unit recurrent layer, another 0.3 dropout layer, and a six-class softmax output layer. The implemented models do not include the 32-unit ReLU dense layer specified in the original proposal.

Both models were trained using the Adam optimizer with a learning rate of 1×10^{-3}, batch size of 32, seed 19, and a maximum of 100 epochs. Early stopping with a patience of 15 epochs and ReduceLROnPlateau were applied during training. Focal loss with $\gamma = 1.5$ was employed together with a class-weighting strategy to address the imbalanced distribution of mental-state classes. Minority-class oversampling with jitter was also incorporated; however, no additional samples were generated in the final training run because the selected classes already exceeded the specified target threshold. Model checkpointing was based on validation macro-F1 to provide balanced consideration across the six mental-state classes.

TABLE II
IMPLEMENTED LSTM/GRU ARCHITECTURES

Layer	**LSTM**	**GRU**
Input	(30, 36)	(30, 36)
Recurrent 1	128 units, L2=10^{-4}	128 units, L2=10^{-4}
Dropout	0.3	0.3
Recurrent 2	64 units	64 units
Dropout	0.3	0.3
Output	Dense 6, softmax	Dense 6, softmax
Parameters	134,278	101,382

The LSTM training terminated at epoch 25, achieving its best validation macro-F1 of 0.4227 at epoch 10. The GRU training terminated at epoch 23, with its best validation macro-F1 of 0.4338 at epoch 8. The corresponding best checkpoints were retained for evaluation on the held-out test set.

E. Equal-Weight LSTM–GRU Ensemble

Softmax probability outputs from the LSTM and GRU classifiers were averaged using equal weights, followed by an argmax operation to obtain the final mental-state prediction[29]-[30]. This approach combines the complementary temporal representations learned by the two recurrent architectures while maintaining a simple ensemble configuration. The two models produced the same prediction for 72.8% of the test sequences, with the shared prediction being correct in 70.0% of these cases. For the remaining 27.2% of sequences where the models disagreed, the LSTM achieved 47.0% accuracy, compared with 29.9% for the GRU and 46.6% for the ensemble. Thus, probability averaging did not improve performance on the disagreement cases compared with the stronger individual classifier. A blend-weight analysis further showed that equal weighting ($\alpha = 0.50$) achieved the best macro-F1 performance, while the highest accuracy was obtained at $\alpha = 0.95$, providing only a 0.21-percentage-point improvement over equal weighting. Therefore, the equal-weight configuration was retained as a simple and consistent ensemble strategy.

F. Rule-Based Decision Agent

The rule-based decision agent converts the predicted pedestrian mental state and time-to-collision (TTC) into an interpretable autonomous-vehicle (AV) action. TTC is calculated from the already extracted feature values. Specifically, the normalized vehicle–pedestrian distance and pedestrian current speed are converted into image-space quantities as $d_{\mathrm{px}} = d_{\mathrm{vehicle}} \times 1920$ and $v_{\mathrm{px}} = v_{\mathrm{current}} \times 1920$, respectively. TTC is then calculated as

$$\mathrm{TTC} = \begin{cases} \dfrac{30 d_{\mathrm{px}}}{v_{\mathrm{px}}}, & v_{\mathrm{px}} > 0, \\ \infty, & v_{\mathrm{px}} \le 0, \end{cases} \tag{1}$$

where 30 represents the video frame rate. Thus, TTC estimates the remaining time to a potential conflict using the pedestrian's current relative image-space motion. The resulting TTC is discretized into three risk levels: **high** ($\mathrm{TTC} < 1.5$ s), **medium** ($1.5 \le \mathrm{TTC} < 2.5$ s), and **low/safe** ($\mathrm{TTC} \ge 2.5$ s or $\mathrm{TTC} = \infty$). This discrete representation provides an auditable alternative to continuous risk functions while preserving the central safety interpretation of TTC.

TTC is widely used as a surrogate safety measure for estimating the temporal proximity of potential collisions [24], [25]. The agent evaluates TTC separately for each detected pedestrian and identifies the pedestrian with the minimum finite TTC as the most critical target. This per-target, worst-case selection is consistent with threat-assessment principles used in forward-collision warning and automated emergency-braking systems [26]. When no pedestrian has a finite TTC, the agent falls back to mental-state severity to select the most urgent pedestrian.

The selected TTC thresholds are motivated by pedestrian-AEB research. The **1.5 s** threshold corresponds to the TTC at which Haus *et al.* [27] reported the largest modeled reduction in pedestrian fatality risk among the evaluated AEB configurations, approximately 84–87%. The **2.5 s** threshold introduces an earlier warning region, providing additional reaction time before the critical TTC threshold is reached and supporting the graded risk interpretation discussed in [28]. Therefore, the thresholds are used as interpretable decision boundaries rather than as learned parameters.

The final AV action is determined solely by the combination of predicted mental state and TTC risk level according to Table III. The implemented agent uses three actions: **MAINTAIN**, **SLOW_DOWN**, and **STOP**. Classifier confidence is recorded for analysis and explanation but does not independently trigger or suppress an action.

For each decision, the agent also generates a predefined natural-language justification, linking the selected action to the pedestrian's mental state and TTC condition. Consequently, the decision process remains deterministic and transparent: **mental state + TTC level → AV action + explanation**.

TABLE III
STATE–TTC LEVEL TO ACTION MAPPING

Mental State	TTC < 1.5 s	$1.5 \le$ TTC < 2.5 s	TTC ≥ 2.5 s / ∞
Waiting	SLOW_DOWN	SLOW_DOWN	MAINTAIN
Hesitant	STOP	SLOW_DOWN	MAINTAIN
Committed	STOP	SLOW_DOWN	SLOW_DOWN
Distracted	STOP	SLOW_DOWN	SLOW_DOWN
Aggressive	STOP	SLOW_DOWN	SLOW_DOWN
Jaywalk	STOP	SLOW_DOWN	SLOW_DOWN

At the frame level, when multiple pedestrians are detected, For each detected pedestrian, the controller independently computes the mental state and TTC using the procedure described above. The resulting pedestrian-level decisions are then compared, and the pedestrian with the minimum finite TTC is selected as the critical target. This most-time-critical-target strategy follows the threat-assessment principle used in AEB/FCW systems [24]-[26]. The final AV action is inherited directly from the selected pedestrian's individual state-TTC decision, while the complete pedestrian ranking and associated decisions are retained for logging and subsequent analysis.

V. EXPERIMENTAL SETUP

Models were implemented in TensorFlow/Keras 2.20. Metrics include accuracy, macro-F1, weighted-F1, and per-class precision, recall, and F1-score using scikit-learn. All classification results are computed on the offline PIE test split. The multi-pedestrian controller was additionally exercised using eleven hand-constructed synthetic scenarios spanning all six mental states and different TTC conditions. These scenarios were used only to verify the decision logic and were not treated as statistical evaluation results.

VI. RESULTS AND DISCUSSION

A. Overall Classification Performance

Table IV summarizes the test-set performance. The LSTM achieves the highest accuracy (63.7% vs. 59.1% for GRU), while the GRU is marginally better in macro-F1 (37.5% vs. 37.1%). The equal-weight ensemble achieves 63.6% accuracy and the highest macro-F1 (38.6%), although the improvement is modest. Compared with the 79% accuracy reported for the binary PIE intention-estimation task [1], the lower performance reflects the greater difficulty of the six-class, pose-based, rule-labeled mental-state classification problem.

TABLE IV
TEST-SET RESULTS

Model	Params.	Acc.	Macro-F1	Wtd.-F1
LSTM	134,278	63.7%	37.1%	65.1%
GRU	101,382	59.1%	37.5%	61.9%
Ensemble	235,660*	63.6%	**38.6%**	65.1%
Target	–	$> 85\%$	$> 80\%$	–

*The ensemble parameter count is the combined count of the two base models; no additional parameters are trained.

B. Per-Class Behavior

Table V shows substantial class imbalance in performance. Waiting is the strongest class, with F1 scores of 0.74–0.79, followed by Aggressive, with 0.65–0.67. Committed and Jaywalk achieve moderate but substantially lower F1 scores, while Distracted and Hesitant remain difficult to classify. Hesitant is the weakest class, with F1 scores of 0.04–0.10.

The low Hesitant performance is not equivalent to an absence of learned signal. The mean predicted probability for Hesitant is higher for true-Hesitant samples than for other classes, but it remains too small to consistently become the maximum predicted probability. Committed also remains challenging despite its larger support, suggesting that class frequency alone does not explain its errors. The ensemble

TABLE V
PER-CLASS F1 (LSTM / GRU / ENSEMBLE)

Class	N	LSTM	GRU	Ens.
Waiting	1,741	0.79	0.74	0.78
Hesitant	59	0.04	0.10	0.10
Committed	326	0.32	0.35	0.33
Distracted	78	0.10	0.13	0.13
Aggressive	412	0.65	0.65	**0.67**
Jaywalk	175	0.32	0.28	0.29

produces an intermediate Committed F1, indicating that probability averaging does not fully resolve disagreement between the two recurrent models.

C. Model Size, Latency, and Prior Work

The GRU uses 24.9% fewer parameters than the LSTM (101,382 vs. 134,278) and is also faster under CPU-only batched inference. With batch size 32, average inference time was 0.950 ms/sequence for the LSTM and 0.397 ms/sequence for the GRU, both below the project's 100-ms target. These measurements represent batched CPU inference rather than single-sample or target-hardware latency.

Prior PIE-derived studies using richer inputs or simpler formulations report substantially higher performance, including 98.3% [13] and 92% [12], compared with 63.7% accuracy in this work. The LSTM–GRU ordering is also mixed: the LSTM achieves higher accuracy, whereas the GRU achieves slightly higher macro-F1. This suggests that relative model performance is configuration-dependent rather than consistently favoring one recurrent architecture[11].

D. Rule-Based Agent Behavior

Eleven synthetic scenarios were used to verify the implemented decision rules across all six mental states and different TTC levels. The results confirm that the decision logic follows the specified state–TTC mapping, with more critical TTC conditions producing more conservative actions. In particular, Waiting does not trigger STOP in the current rule table; its most conservative response is SLOW_DOWN. This represents a deliberate rule-design choice and should be reconsidered before making any safety-performance claims.

VII. LIMITATIONS AND FUTURE WORK

Model performance: The current classification performance remains below the desired targets. Further hyperparameter tuning, architecture optimization, and improved sequence modelling could increase accuracy and macro-F1.

Class imbalance: Minority classes, particularly Hesitant and Distracted, remain difficult to classify. Increasing representative samples and applying class-specific augmentation or sampling strategies could improve their recognition.

Feature representation: The current model relies on the extracted 36-feature representation. Incorporating additional temporal, contextual, gaze, and pedestrian–vehicle interaction features could improve discrimination between similar mental states.

Label quality: The mental-state labels are derived from rule-based interpretations of PIE annotations. Additional manually validated annotations could provide more reliable supervision and improve model robustness.

Ensemble optimization: The current ensemble uses equal weighting between LSTM and GRU predictions. Optimized or class-specific weighting could potentially improve ensemble performance.

Decision-agent refinement: The current decision agent uses predefined mental-state and TTC rules. Adaptive TTC thresholds and more flexible decision policies could improve its behaviour under different traffic conditions.

Evaluation: The decision logic was verified using eleven synthetic scenarios. A larger and more diverse evaluation set would provide stronger evidence of robustness across different pedestrian behaviours and traffic conditions.

Deployment: Inference latency was measured using CPU-only batched inference. Future evaluation on target hardware and under real-time single-sample conditions would provide a more representative assessment of deployment efficiency.

VIII. CONCLUSION

This paper presented a fully implemented pipeline for pedestrian mental-state inference and explainable AV decision-making, integrating corrected pose-based features, rule-based six-class labeling, two temporal classifiers, an equal-weight LSTM–GRU ensemble, and a TTC-based rule-driven decision agent for multiple pedestrians. The LSTM, GRU, and ensemble achieved accuracies of 63.7%, 59.1%, and 63.6%, with macro-F1 scores of 37.1%, 37.5%, and 38.6%, respectively. The ensemble achieved the highest macro-F1, while the LSTM achieved the highest accuracy. The results demonstrate the feasibility of combining temporal mental-state inference with interpretable TTC-based decision rules, while also highlighting opportunities for improving minority-class recognition, feature representation, and decision robustness.

ACKNOWLEDGMENT

The authors sincerely appreciate the guidance, encouragement, and constructive feedback provided by their supervisors throughout this research. Their expertise and continuous support were instrumental in shaping and refining this work. The authors also extend their gratitude to everyone who provided valuable assistance, suggestions, and encouragement during the completion of this study.

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